

Probabilistic Roadmap Planning for a Planar Three-Link Robot

CENG 786 - Robot Motion Planning and Control

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1. Introduction

Path planning is a cornerstone of robotics and automation, involving the determination of feasible trajectories for robots to navigate environments filled with obstacles. Among the diverse path planning strategies, **Probabilistic Roadmap (PRM)** algorithms stand out for their robustness in solving motion planning problems in high-dimensional configuration spaces[2]. PRMs construct a graph or "roadmap" of viable paths, enabling robots to traverse from a start configuration to a goal configuration efficiently.

The PRM framework was first proposed by **Kavraki et al. in 1996**[1], introducing a two-phase approach: the roadmap construction phase and the query phase. During the construction phase, the algorithm samples the configuration space (C-space) to identify collision-free configurations and establishes connections between them based on their feasibility. The query phase utilizes the roadmap to find a path from the start to the goal configuration, often employing graph search algorithms such as A*. Extensions and adaptations of PRMs have since addressed various challenges in motion planning, including environments with narrow passages and dynamic obstacles[5].

Theoretical foundations of sampling-based approaches, such as those laid out by **LaValle's Planning Algorithms**[2], provide a broader context for PRM's applications. Innovations like the **Single-Query Bi-Directional PRM**[7] and **Lazy PRM**[10] have introduced efficiency improvements by minimizing unnecessary sampling or collision checks. More recent advancements, including **Learned Critical PRMs**[3], leverage machine learning to optimize sample placement, while **SwarmPRM**[4] extends the PRM framework to large-scale multi-agent systems.

PRMs have shown versatility in diverse applications, from motion planning in virtual environments[9] to optimizing paths for robotic manipulators in cluttered workspaces[8]. Their capability to adapt to high-dimensional spaces and irregular environments makes them indispensable in both theoretical and practical robotics.

This project focuses on implementing a PRM for a planar robot with three links and two joints, a

common robotic architecture[6]. By simulating various scenarios, including narrow passages and cluttered obstacle fields, the project evaluates the strengths and limitations of the PRM algorithm. Key performance factors, such as obstacle density, connection distance, and computational cost, are analyzed to provide insights into optimizing PRMs for real-world applications[11].

2. Problem Definition

Problem Definition

Motion planning is a critical component of robotic systems, enabling robots to navigate through complex environments while avoiding obstacles and achieving specific goals. The Probabilistic Roadmap (PRM) algorithm is a widely used sampling-based motion planning technique that provides an efficient solution for high-dimensional spaces. However, its performance is heavily influenced by the configuration of the environment, the nature of the robot, and the parameters chosen for sampling and connection.

In this project, the focus is on utilizing the PRM algorithm for a planar robot with three rigid links connected by two revolute joints. The robot must navigate through various challenging scenarios, such as narrow passages and environments with densely placed obstacles. These scenarios test the algorithm's capability to efficiently sample valid configurations, build a collision-free roadmap, and find feasible paths between a start and a goal configuration.

Specific challenges arise in:

1. **Collision Checking:** Ensuring that the robot, with its multiple links and joints, avoids all obstacles, including narrow passages.
2. **Scalability:** Managing computational efficiency when the number of obstacles or sampling points increases significantly.
3. **Parameter Sensitivity:** Balancing the number of samples, connection distance, and other parameters to achieve a practical trade-off between computational time and path quality.
4. **Validation of Results:** Ensuring that the paths generated are feasible for the robot's kinematics and are collision-free in all configurations.

By addressing these challenges, this project aims to:

- Enhance understanding of PRM's implementation and behavior under different environmental conditions.
- Evaluate the algorithm's performance with a planar robot with three links in a variety of test scenarios.
- Identify potential improvements to increase efficiency and robustness in motion planning tasks.

This study provides insights into the adaptability and limitations of PRM for high-dimensional and constrained environments, contributing to its application in real-world robotic systems.

3. Methodology

Environment and Robot Definition

The workspace is a bounded 2D plane with user-defined obstacles. Obstacles are represented as polygons, and narrow passages are created by strategically placing obstacles close to each other. The planar robot is defined as having three rigid links and two revolute joints. Each configuration of the robot includes:

- The base position (x, y) ,
- Joint angles $(\theta_1, \theta_2, \theta_3)$.

The workspace boundaries and obstacles are stored as geometric primitives, while the robot's kinematics are represented mathematically to compute link positions.

Roadmap Construction

The PRM algorithm builds a graph where nodes represent collision-free configurations, and edges represent feasible transitions between these configurations. The process includes:

1. **Sampling Valid Configurations:**

- Random configurations of the robot are generated within the workspace boundaries.
- Each configuration is validated to ensure all robot links avoid obstacles using a collision-checking function.

2. **Connection of Configurations:**

- For each sampled configuration, its K -nearest neighbors are determined using a distance metric.
- Feasibility of a direct path between configurations is validated using incremental collision checking.

3. **Graph Representation:**

- The configurations are stored as graph nodes, and valid connections between them are stored as graph edges with weights proportional to the Euclidean distance.

Pathfinding

Once the roadmap is constructed, a path from the start to the goal configuration is computed using the A* algorithm. Key steps include:

- Defining a heuristic function based on the Euclidean distance to the goal.
- Traversing the roadmap to find the shortest collision-free path.

Pseudo Code for Pathfinding:

```
function AStar(graph, start, goal, heuristic):  
    open_set = PriorityQueue()  
    open_set.insert(start, 0)  
    came_from = {}  
    g_score[start] = 0
```

```

f_score[start] = heuristic(start, goal)

while open_set is not empty:
    current = open_set.extract_min()
    if current == goal:
        return reconstruct_path(came_from, current)

    for neighbor in graph.neighbors(current):
        tentative_g_score = g_score[current] + graph.cost(current,
neighbor)
        if tentative_g_score < g_score.get(neighbor, infinity):
            came_from[neighbor] = current
            g_score[neighbor] = tentative_g_score
            f_score[neighbor] = tentative_g_score +
heuristic(neighbor, goal)
            if neighbor not in open_set:
                open_set.insert(neighbor, f_score[neighbor])

return failure

```

Collision Checking

Collision checking involves:

- Computing the positions of all robot links for a given configuration using forward kinematics.
- Verifying that none of the links intersect with any obstacles in the workspace.

Testing and Evaluation

The algorithm is tested under various scenarios, including:

- Environments with narrow passages.
- Increasing obstacle density.
- Configurations with varying sampling densities and connection distances.

Metrics such as computation time, path quality, and success rate are used to evaluate performance. Scenarios are visualized to analyze the paths generated for all robot links.

Visualization

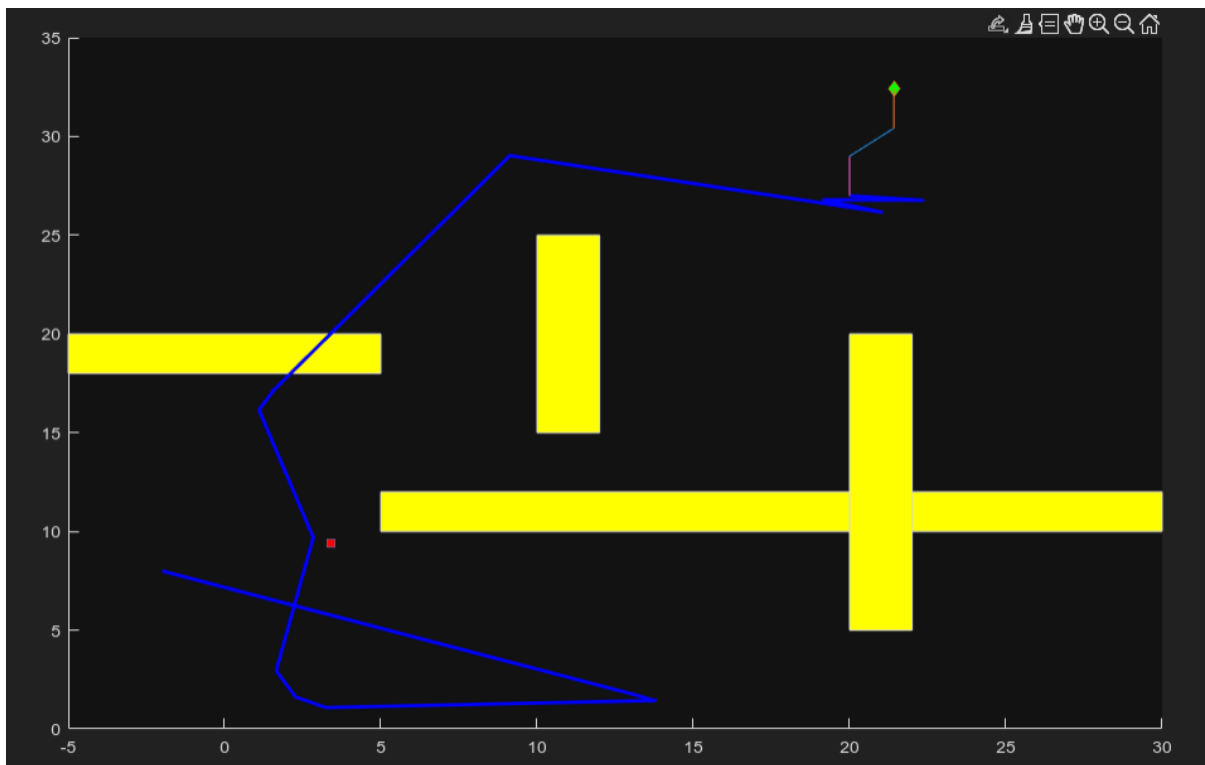
The paths of the robot's links are plotted over the workspace for visual analysis. Videos of the robot's motion along the computed path are generated to demonstrate the feasibility of the solution.

4. Experiments and Results

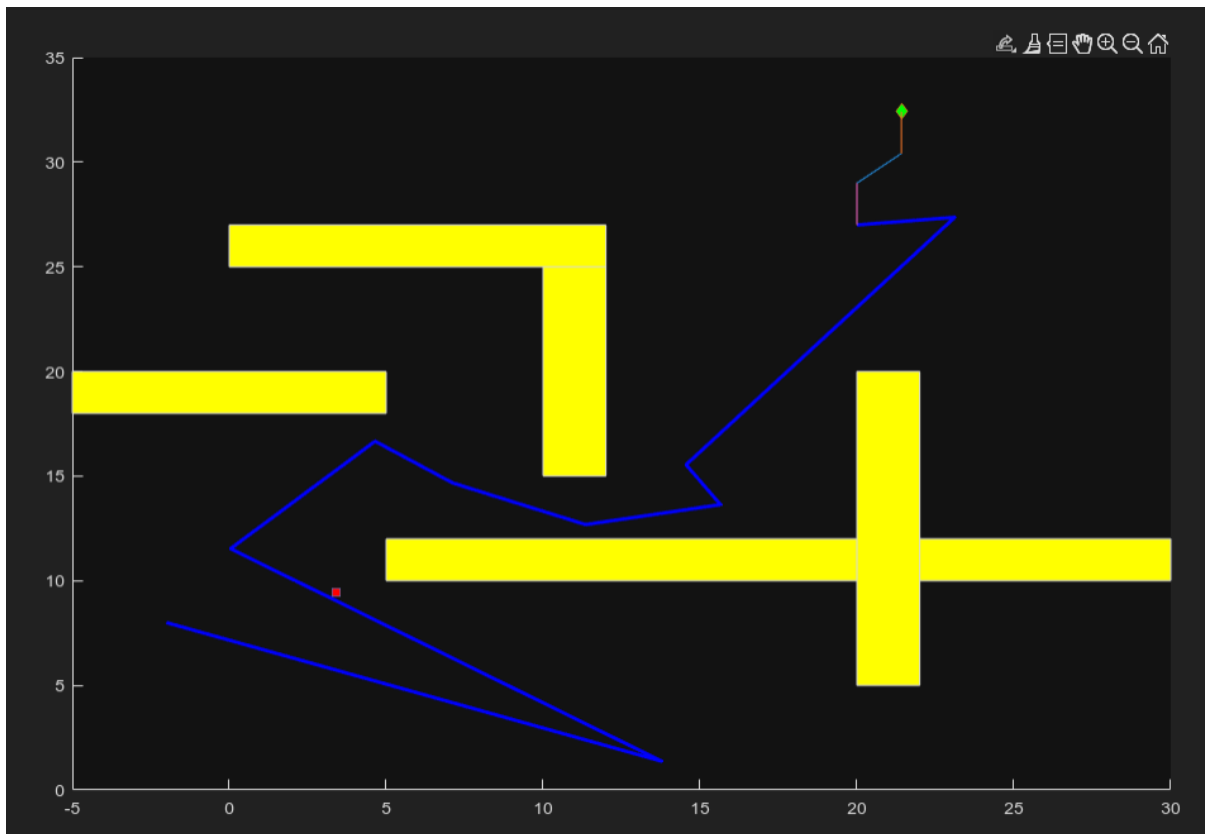


Fig(1) This behavior demonstrates the characteristic randomized nature of the PRM algorithm, which can result in non-optimal paths with zigzag motions through the configuration space.

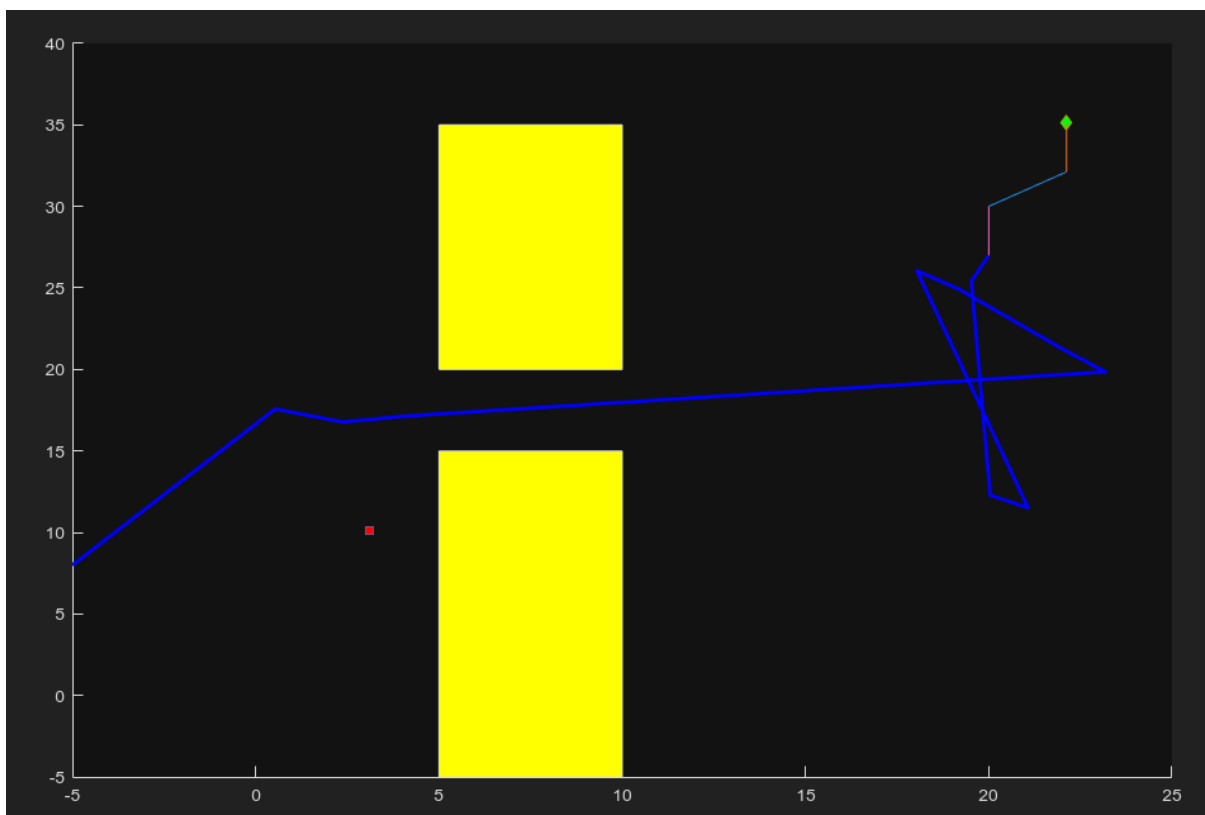
Fig(3) In this experiment, the robot successfully navigated through the maze environment to reach the goal configuration without any collisions. The PRM algorithm was configured with 1000 random samples and a connectivity parameter of 10, while each robot link was set to a length of 2 units.



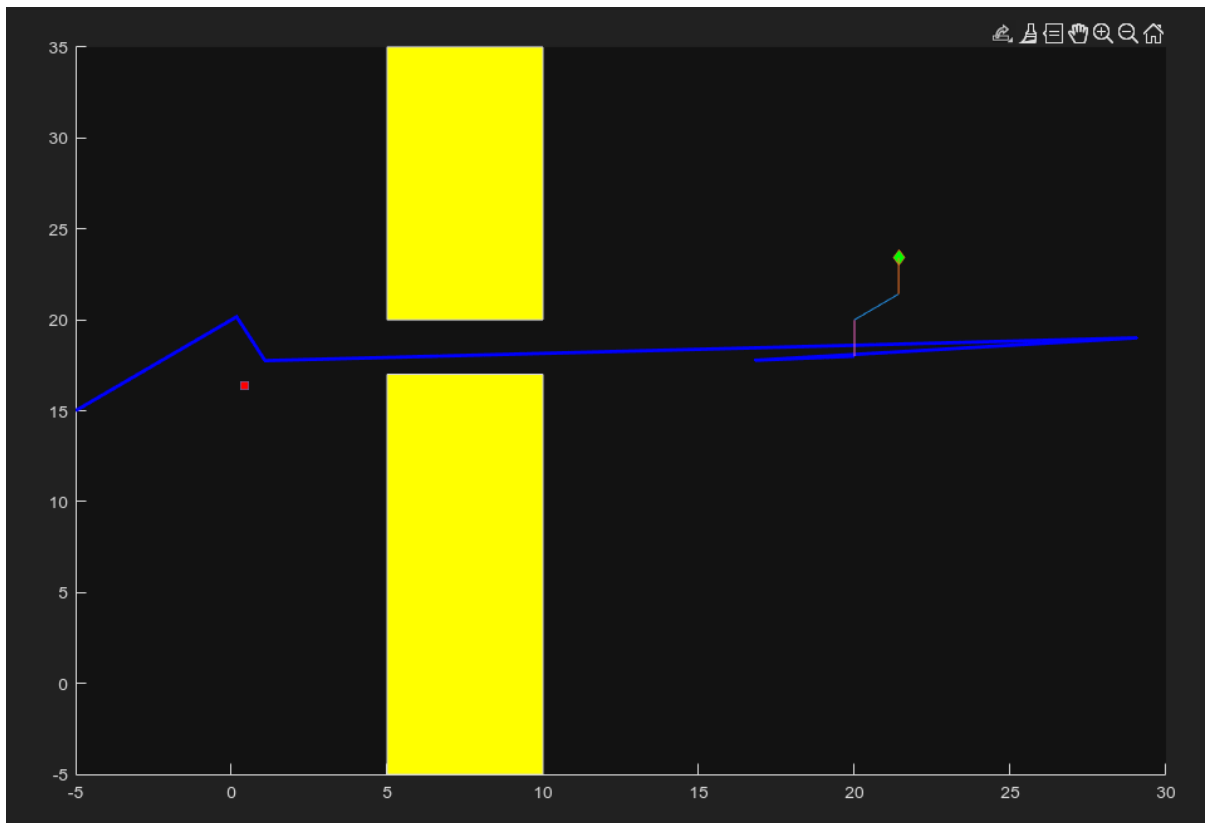
Fig(4) The algorithm's performance was impacted by the relationship between obstacle thickness and connectivity parameters. Despite the obstacle thickness being less than the connectivity length, the discretization of collision checking into 21 steps resulted in a sampling resolution that exceeded the passage width. This relationship between sampling granularity and geometric constraints significantly influenced the algorithm's ability to find valid paths through narrow passages.



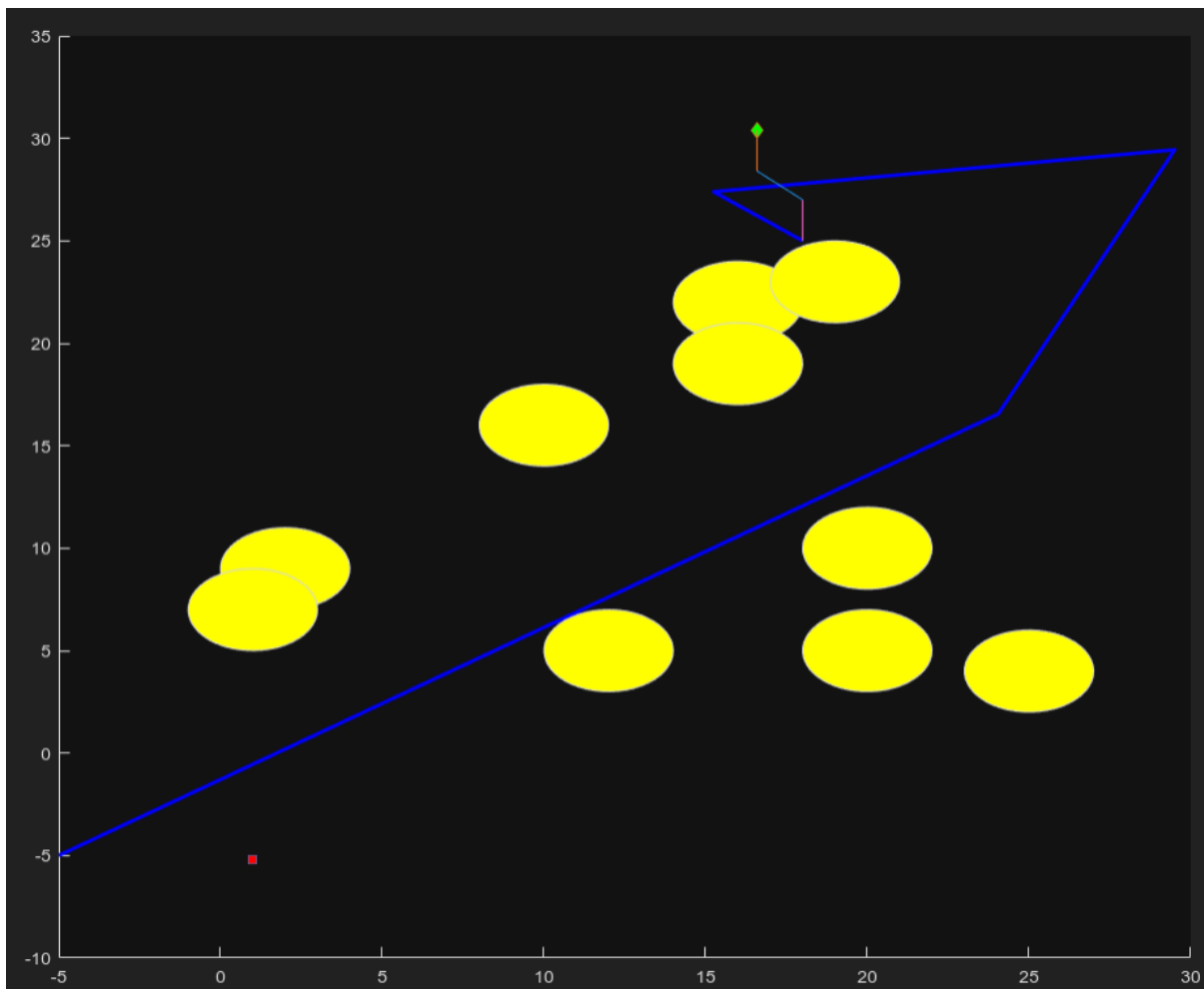
Fig(5) In this iteration, the algorithm's performance was enhanced through increased sampling resolution, resulting in improved path planning capabilities through narrow passages.



Fig(6) The robot successfully navigated through a narrow passage with a width of 5 units, demonstrating the PRM algorithm's capability to find valid paths in constrained environments.

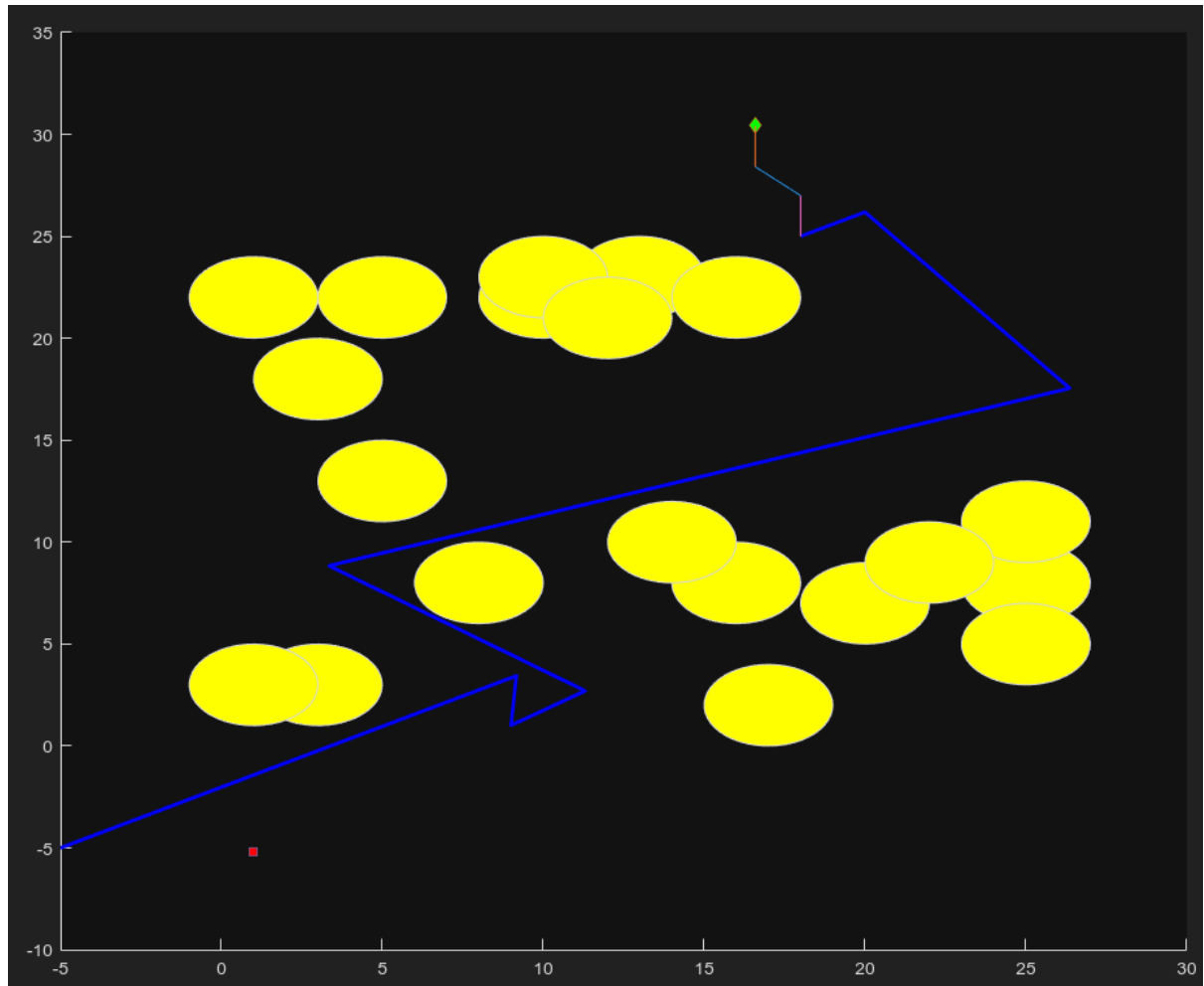


Fig(7) The robot successfully navigated through a narrow passage with a width of 3 units, demonstrating the algorithm's effectiveness in highly constrained environments.



Fig(8) The workspace environment was populated with 10 randomly distributed circular

obstacles to evaluate the algorithm's performance in scenarios with scattered collision regions.



Fig(9) The workspace environment was populated with 20 randomly distributed circular obstacles to evaluate the algorithm's performance in scenarios with increased obstacle density and more complex collision regions.

1. Experiment Setup

The experiments were conducted to evaluate the performance of the Probabilistic Roadmap (PRM) algorithm in different test scenarios. The planar robot consists of three rigid links connected by two revolute joints, operating in a 2D workspace with various obstacle configurations. The start and goal configurations were fixed for each scenario, and the results were analyzed in terms of computation time, path feasibility, and algorithm performance.

The scenarios tested include:

- **Scatter Scenario:** Randomly distributed circular obstacles of varying densities.
- **Narrow Pass:** A challenging environment with tightly spaced obstacles forming narrow corridors.
- **Maze Scenario:** A structured environment with walls and predefined passages.
- **Simple Obstacle:** A basic environment with a few simple polygonal obstacles.

The parameters varied across experiments include:

- **Number of random samples (n):** 250 to 2000.

- **Connection distance (K):** 20 to 1500.
- **Step size:** Incremental size for collision checking.
- **Robot link length (r):** Fixed at 5 units but considered for tuning.

2. Results

2.1 Scatter Scenario

- **Description:** Circular obstacles were randomly scattered across the workspace with densities ranging from 10 to 50 obstacles.
- **Findings:**
 - At lower densities (10–20 obstacles), the PRM successfully generated paths with manageable computation times.
 - At higher densities (30+ obstacles), computation time increased significantly due to frequent collision checks.
 - Adjusting the following parameters improved performance:
 - Reducing connection distance to avoid unnecessary connections.
 - Increasing the number of samples (n) to provide better roadmap coverage.
 - Reducing the robot's link length (r) to improve maneuverability.

2.2 Narrow Pass Scenario

- **Description:** Two obstacles created a narrow passage, with the thickness of the passage varying from 2 to 5 units.
- **Findings:**
 - For a passage thickness of 5 units, the algorithm found a path with 1000 steps.
 - For a passage thickness of 2 units, the algorithm initially failed to find a solution due to the limited sampling density and connection feasibility.
 - Increasing the number of steps to 2000 improved success but was computationally expensive.
 - Using smaller robot links or reducing the step size during collision checking improved results for extremely narrow passages.

2.3 Maze Scenario

- **Description:** A structured maze environment with walls forming corridors and dead-ends.
- **Findings:**
 - PRM effectively found paths in the maze with moderate sampling density ($n = 500$) and connection distance ($K = 20$).
 - Increasing the number of samples improved roadmap connectivity and reduced failure rates in highly constrained sections.
 - Computation time remained manageable compared to scatter scenarios.

2.4 Simple Obstacle Scenario

- **Description:** Few polygonal obstacles with ample free space.
- **Findings:**
 - The algorithm performed efficiently with low computation time and high success rates.
 - Even with reduced samples ($n = 250$) and connection distance ($K = 10$), PRM

successfully planned paths.

3. Analysis

- **Impact of Obstacle Density:** Higher obstacle densities significantly increased computation time due to the complexity of collision checks. Reducing connection distances and increasing sampling density provided better roadmap coverage.
- **Narrow Passes:** Tuning parameters such as step size and robot dimensions were critical for solving narrow passage scenarios. Computational expense increased for smaller passages, highlighting the algorithm's sensitivity to geometric constraints.
- **Overall Feasibility:** PRM demonstrated robust performance across varying scenarios, with success rates heavily dependent on parameter tuning and scenario complexity.

4. Visualization

- For each scenario, paths for all robot links were plotted alongside the workspace and obstacles. The visualizations helped identify problematic areas in the roadmap and provided insights into the algorithm's performance.
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5. Discussion

The experiments conducted highlight the strengths and limitations of the Probabilistic Roadmap (PRM) algorithm in various planning scenarios. PRM proved to be a flexible and effective method for path planning, particularly in environments with moderate obstacle densities and relatively simple geometric constraints. However, several key insights emerged from the study:

1. Strengths of PRM:

- **Scalability:** PRM handles diverse environments well by incrementally building a roadmap, making it suitable for large workspaces.
- **Adaptability:** By tuning parameters such as sample size, connection distance, and step size, PRM can adapt to varying complexities and constraints.
- **Visualization:** The clear visualization of roadmaps and paths provides valuable insights into algorithm performance and configuration space coverage.

2. Challenges:

- **High Obstacle Density:** As seen in the scatter scenario, increased obstacle density results in a dramatic rise in computation time due to the number of collision checks required. This poses scalability challenges for highly cluttered environments.
- **Narrow Passages:** PRM struggles in scenarios with narrow passages, as seen in the narrow pass test. Increased sampling density and reduced robot dimensions improved performance but at the cost of computational resources.
- **Parameter Sensitivity:** PRM's performance is heavily dependent on appropriate parameter selection. Parameters such as step size, connection distance, and number of samples significantly influence success rates and computation time.

3. Insights for Improvement:

- Incorporating hybrid approaches, such as combining PRM with local planners or heuristics, can improve success rates in narrow passage scenarios.
 - Adaptive sampling strategies, where more samples are concentrated in challenging regions, can enhance roadmap coverage without significantly increasing computational demands.
 - Exploring alternative collision-checking methods or using parallel processing for collision detection can mitigate the computational bottlenecks observed in dense environments.
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6. Conclusion

The conducted experiments validate the efficacy of the PRM algorithm in planning paths for a planar 3-link robotic manipulator across diverse scenarios. PRM exhibited robust performance in environments with moderate complexity, while challenges such as high obstacle density and narrow passages highlighted the need for careful parameter tuning and potential algorithmic enhancements.

The study demonstrates that PRM is a viable approach for motion planning, particularly for robots with fixed link numbers and joint configurations. However, future work should focus on addressing its limitations, optimizing computational efficiency, and incorporating hybrid methods to enhance its applicability in real-world, complex environments.

7. References

1. Kavraki, L. E., Švestka, P., Latombe, J.-C., & Overmars, M. H. (1996). Probabilistic roadmaps for path planning in high-dimensional configuration spaces. *IEEE Transactions on Robotics and Automation*, 12(4), 566–580. (<https://ieeexplore.ieee.org/document/508439>).
2. LaValle, S. M. (2006). *Planning Algorithms*. Cambridge University Press. (<http://planning.cs.uiuc.edu/>).
3. Ichter, B., Schmerling, E., Lee, T.-W. E., & Faust, A. (2019). Learned critical probabilistic roadmaps for robotic motion planning. *arXiv Preprint arXiv:1910.03701*. (<https://arxiv.org/abs/1910.03701>).
4. Hu, Y., Yang, X., Zhou, K., et al. (2024). SwarmPRM: Probabilistic roadmap motion planning for large-scale swarm robotic systems. *arXiv Preprint arXiv:2402.16699*. (<https://arxiv.org/abs/2402.16699>).
5. Amato, N. M., & Wu, Y. (1996). A randomized roadmap method for path and manipulation planning. *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*. (<https://ieeexplore.ieee.org/document/508494>).
6. Elbanhawi, M., & Simic, M. (2014). Sampling-based robot motion planning: A review. *IEEE Access*, 2, 56-77. (<https://ieeexplore.ieee.org/document/6807764>).

7. Sanchez, G., & Latombe, J.-C. (2002). A single-query bi-directional probabilistic roadmap planner with lazy collision checking. *International Journal of Robotics Research*, 21(1), 5-26. (<https://journals.sagepub.com/doi/10.1177/0278364902021001956>).
8. Overmars, M. H., & Svestka, P. (1995). A probabilistic learning approach to motion planning. *Algorithmica*, 14(6), 469-494. (<https://doi.org/10.1007/BF01294163>).
9. Geraerts, R., & Overmars, M. H. (2007). Creating high-quality roadmaps for motion planning in virtual environments. *IEEE Transactions on Visualization and Computer Graphics*, 13(5), 812-825. (<https://ieeexplore.ieee.org/document/4287621>).
10. Bohlin, R., & Kavraki, L. E. (2000). Path planning using lazy PRM. *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*. (<https://ieeexplore.ieee.org/document/844730>).
11. Hsu, D., Latombe, J.-C., & Motwani, R. (1999). Path planning in expansive configuration spaces. *International Journal of Computational Geometry & Applications*, 9(4), 495-512. (<https://doi.org/10.1142/S0218195999000292>).
12. Karaman, S., & Frazzoli, E. (2011). Sampling-based algorithms for optimal motion planning. *International Journal of Robotics Research*, 30(7), 846-894. (<https://journals.sagepub.com/doi/10.1177/0278364911406761>).